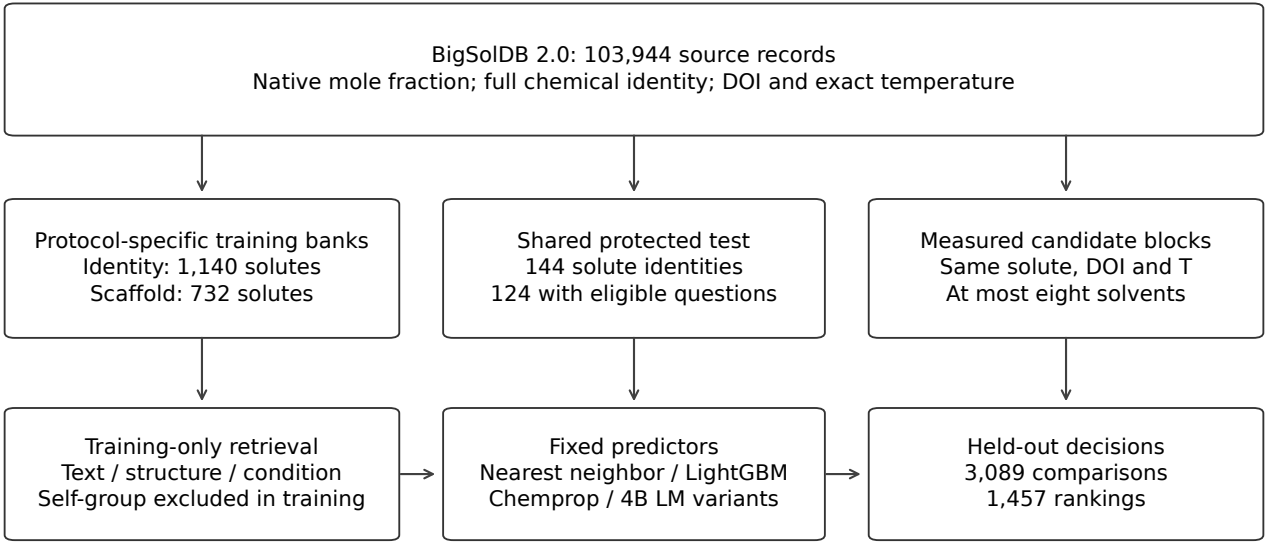
BigSolDB 2.0: 103,944 source records
Native mole fraction; full chemical identity; DOI and exact temperature
Protocol-specific training banks
Identity: 1,140 solutes
Scaffold: 732 solutes
Shared protected test
144 solute identities
124 with eligible questions
Measured candidate blocks
Same solute, DOI and T
At most eight solvents
Training-only retrieval
Text / structure / condition
Self-group excluded in training
Fixed predictors
Nearest neighbor / LightGBM
Chemprop / 4B LM variants
Held-out decisions
3,089 comparisons
1,457 rankings
Solute-macro scores; matched-seed contrasts; option-order and equivalent-SMILES checks